



Big Data Laboratory – Team Project

June 3rd , 2026, Bologna

Maximum team size: 4 students

Overview

The team project involves two distinct tasks: one focused on **computer vision** and the other on **large language model** (LLM) applications. These tasks are designed to assess your ability to implement, analyze, and critically evaluate machine learning solutions in different domains.

Each team is **required to choose and complete at least one** of the two tasks with a focus on demonstrating deep understanding rather than achieving perfect accuracy scores. While we expect your models to produce reasonable and meaningful results, the most important aspect is your comprehension of the underlying concepts, methodologies, and decision-making processes for each individual task.

During the practical implementation, **review the relevant lecture materials** to refresh your understanding of the theoretical foundations. This will help you make informed decisions throughout the project and articulate your reasoning during the presentation. Additionally, you are encouraged to consult **official documentation and resources** for the libraries and frameworks you'll be using since it will provide valuable technical guidance for implementation details.

Material:

To get the material for the team project, please use:

git clone https://gitlab.hpc.cineca.it/rmioli00/hpc_bbs_26.git

in order to gain access to all the lessons material. If you already have cloned the repository in your working directory, just run “git pull” to update the material. Now, inside the directory “team_project” you’ll find two directories, one for each project. Both of them contains a description of the tasks you are supposed to tackle.

We have prepared **two notebooks**, **student_notebook_CV.ipynb** and **student_notebook_LLM.ipynb** to guide you through the two tasks. The first is more structured, while the second leaves more space for experimentation. You can start with the one you prefer.

You can find the two notebooks and the **two job scripts CV_start_jupyter.sh** and **LLM_start_jupyter.sh** to start them, respectively, here: /leonardo_work/tra26_bbs/hpc_bbs_26/team_project. Copy it to your folder as we’ve done during the lectures.

Specific project instructions are available in the next few sections.

Final presentation:

During your final presentation, you will be expected to demonstrate comprehensive understanding through:

- Materials to present:
 - **Jupyter Notebooks** (minimum one of the two tasks completed)
 - **Results analysis** including visualizations and performance metrics for each task
 - **Methodology explanation** detailing your design choices for each task
 - **Critical evaluation** of your work for each task, including limitations and areas for improvement
- Discussion topics:
 - Your implementation choices and reasoning
 - Challenges encountered and how you addressed them
 - Weak points in your approach and potential solutions
 - Alternative methodologies you considered
 - Improvements that could enhance your results
 - Individual contributions within the team
- Answering and discussing the questions presented in the Jupyter Notebooks.

Task 1 (Computer vision) - Guardian AI: smart environmental surveillance

Background

Governments and NGOs are collaborating on a project called Guardian AI, which aims to automate environmental monitoring in protected areas using drones and camera traps. These AI-powered agents must classify the environment, detect objects of interest within the scene, and generate alerts for illegal activity or anomalies.

Your team is tasked with building the scene classification component of Guardian AI's core vision system.

Objective

Scene Classification: Classify the overall setting of each image to understand the type of environment being monitored (e.g., urban, forest, desert, coastal, agricultural, industrial).

Technical Requirements

To achieve this objective, you should:

- First, train a CNN from scratch as described in the notebook,
- Then, use an existing architecture (e.g., ResNet18).

Note that in order to obtain a minimum score of 18, at least the implementation of the training from scratch should be presented.

Dataset

Places365

- A scene recognition dataset with images labeled into 365 scene categories
- Features: 1.8 million training images and 36,500 validation images
- The dataset is available through [TensorFlow Datasets](#) and [PyTorch's torchvision](#)
- List of available categories:
https://github.com/zhoubolei/places_devkit/blob/master/categories_places365.txt

The data has already been downloaded and can be found in the folder: `/leonardo_scratch/fast/tra26_bbs/data/places365/`.

Note: You should decide which specific scene categories are most relevant for environmental monitoring and focus on your classification system accordingly.

Task 2 (LLM) - The LLAMA of WallStreet: LLM data extraction and sentiment analysis

Background

You work as Data Scientist in a company which develops algorithmic trading strategies. Your supervisor aims to develop a sentiment analysis pipeline for Reddit posts, with the objective of incorporating Reddit comments into an automated trading strategy. The goal is to leverage the content of these comments to extract actionable insights for identifying potential stock buying or selling opportunities.

Objectives

1. **Company identification:** each comment you are going to process must be associated with one or more stock tickers if the comment is about a company traded on a stock exchange. You can perform this “data extraction task” using an LLM.
If the comment is not relevant, no ticker should be associated with the comment.
2. **Sentiment extraction:** Each comment about a company must be classified as Very Positive, Positive, Neutral, Negative, Very Negative. You can extract the sentiment using an LLM or a model built specifically for this task (e.g. <https://huggingface.co/cardiffnlp/twitter-roberta-base-sentiment-latest>).

Material

In the folder of the project you'll find:

1. README.md: describes the project's tasks, has some suggestions and initial configurations
2. reddit_comments.csv: the dataset, described in next section
3. requirements.txt: a starting point for your python environment, but you can install any other package you may want/need.
4. student_notebook_LLM.ipynb: A notebook containing an example of how to classify a comment
5. Student job LLM.py: python script where you should write your final solution

Dataset

The dataset consists of a CSV file with 100k comments. The columns are organized in the following way:

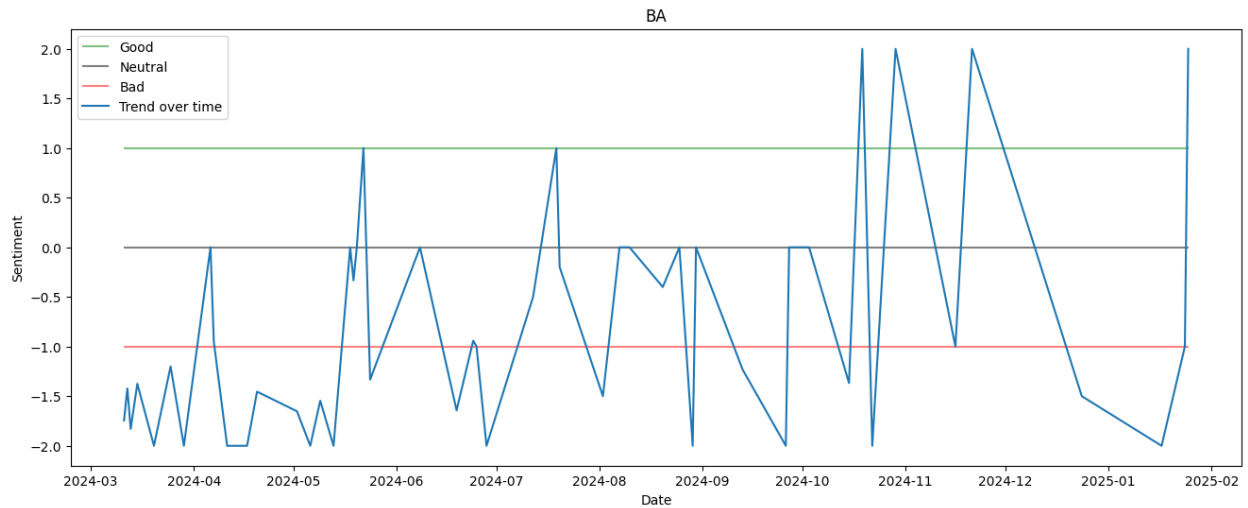
- **datetime**: the datetime associated to a comment;
- **subreddits**: the subreddit in which the comment was posted;
- **submission_id**: an id associated to the subreddit post;
- **comments**: the comment to be processed.

Implementation flow

The pipeline you are going to create does not have any particular constraint about the technologies to be used; however, given the moderate number of comments to be processed, it is strongly advised to parallelize your workflow in some way.

1. Extract one or more tickers from each comment. To perform this task, you should use an LLM. A mistral 3.2 24b will be loaded for you by the jobscripsts associated to this assignment;
2. Associate a sentiment (very positive, positive, neutral, negative, very negative) to each comment. To perform this task you can use an LLM or a different kind of model trained specifically for this task;
3. Map sentiment to integers: 2, 1, 0, -1, -2;
4. Once you have mapped sentiment to numbers, calculate a daily sentiment for each ticker;
5. Calculate some metrics (e.g. min, max, average sentiment) for each ticker on a daily basis;
6. Plot the sentiment trend for some tickers you find interesting (e.g. AAPL, META, etc.)

7. *Design* a plausible implementation of a SLURM agent, capable of using sbatch and queue commands to start jobs and check their execution.
8. (optional) Implement the SLURM agent you designed.



An example of the expected output. The company in the screenshot is Boeing Co.
(company ticker is BA).